

Disentangling the Effects of Social and Emotional Learning Programs on Student Academic Achievement Across Grades 1–12: A Systematic Review and Meta-analysis

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Universal school-based social and emotional Learning (SEL) programs are designed to provide structured learning opportunities for all students during the regular school time to advance their social, emotional, and academic development. This systematic review and meta-analysis reviewed 40 empirical studies ($n = 40$) conducted between 2008 and 2020 and explored the effect of SEL programs on students' academic achievement across first through twelfth grade. The analysis involved 33,737 students and 76 effect sizes ($k = 76$) related explicitly to academic achievement. Findings reveal that students in grades 1–12 who participated in universal school-based SEL programs demonstrated better academic achievement ($g = .101$) compared to those who did not participate. Analyses further confirmed that student standardized test scores and grade point averages are improved by participating in SEL programs and that student test scores are the most proximal and accurate estimate of the effects of an SEL program on student achievement. Furthermore, students showed significantly improved achievement in both literacy and math across grades 1–12, suggesting the impact of SEL across academic domains. Implications for addressing the academic achievement gap through SEL programs are discussed.

KEYWORDS: social and emotional learning (SEL), child and adolescent development, meta-analysis, systematic review, achievement

Youth worldwide currently face a multitude of challenges that are impacting their academic achievement. Most notable is the lingering impact of the COVID-19 pandemic, which resulted in lost learning and spurred chronic absenteeism among students across the globe. In 2022, the Organization for Economic Cooperation and Development (OECD) published data pertaining to their Programme for International Student Assessment (PISA), a global report that examines academic data related to math, science, and reading among 15-year-olds. Results indicated that since the last study published in 2018, there was an average drop of 10 points in reading and 15 points in math (OECD, 2023). This is particularly concerning given that there has never been more than a five-point change in reading and a four-point change in math between PISA studies (OECD, 2023).

When considering ways to improve academic achievement in areas like math and reading, one obvious approach is to use programs that target specific subject matters. However, many factors influence student academic outcomes that are not directly addressed by academic intervention alone. For example, research has shown that students who feel a sense of belonging within the school community are more successful academically (Korpershoek et al., 2019), yet absenteeism data suggest students are becoming less connected to and engaged in school (Dee, 2024). Additionally, students worldwide face disparate educational opportunities—a reality with substantial effects on academic performance (Parker et al., 2018). OECD recently reported that students with low socioeconomic status scored 93 points less, on average, than their socioeconomically advantaged peers in mathematics (OECD, 2023). Coupled with learning losses and inequitable learning opportunities, nearly 90 million boys and 80 million girls ages 10 to 19 are living with a mental health disorder, with more than a third being diagnosed with anxiety and depression (UNICEF, 2021). Sense of belonging and pervasive mental health challenges only represent a portion of the factors that can impact student academic success. Thus, schools are increasingly being confronted with a variety of factors that are adversely impacting student academic achievement, which requires additional social and emotional support.

Researcher indicates that students' social, emotional, and cognitive development is strongly connected and integral to students' effective learning and positive academic outcomes (Immordino-Yang, 2011; Immordino-Yang et al., 2019; S. M. Jones & Zigler, 2002; S. Jones et al., 2017). Indeed, there is a growing consensus that schools improve a range of student outcomes when focusing explicitly on social and emotional skill development concurrently with academic skills (Cook et al., 2021; Schonfeld et al., 2015; Singh & Duraiappah, 2020). Although children and youth also develop social-emotional competencies (SECs) from their everyday life situations (Elias & Haynes, 2008; Greenberg et al., 2003), they widely vary in their opportunities for exposure to such experiences that benefit their outcomes. If the goal of schooling is to support the academic achievement of all students to reach their fullest potential, then providing all students with the opportunities to access experiences that improve SECs is vital to this process (Cipriano & McCarthy, 2023).

Universal SEL: What Is It and Why Is It Important for Schools?

SEL programs are organized to improve social, emotional, behavioral, and mental health by developing students' intra- and inter-personal skills (*see* Cipriano, Naples, et al., 2023a; Cipriano, Strambler, et al., 2023b; S. Jones et al., 2017). The primary feature of universal SEL programs is that they are delivered to all students in a specific classroom, grade, school, or school district rather than solely targeting the social and emotional needs and/or risks of a specific group of learners (e.g., only boys). SEL targets a wide range of competencies and skills that include behavioral regulation, identity development, and emotional intelligence and can be beneficial to the culture, climate, and community of a school by developing skills among students that promote and sustain supportive relationships as well as more equitable and empathetic decision-making (Cipriano, Naples, et al., 2023a; Cipriano, Strambler, et al., 2023b).

In terms of educational equity, accessible SEL contributes to ensuring that all school students have opportunities for their social and emotional development (Cipriano, Naples, et al., 2023a; Cipriano & Strambler, 2024). Particularly, universal SEL programs have been offered to all students in a school within classroom environments, which also means that some learners are not excluded on the surface from this educational access for specific reasons (e.g., students with disabilities) (Greenberg et al., 2003; Rappolt-Schlichtmann et al., 2013; Rivers et al., 2013). Therefore, as a feasible and equitable educational approach, universal school-based SEL can help young learners reach their holistic developmental potential (Osher et al., 2020). Reviewing recent SEL research and exploring evidence synthesized specifically for academic achievement can provide important guidance for SEL decision-making.

Previous SEL Meta-Analysis Findings

Previous meta-analyses of SEL have indicated the positive influence of universal SEL programs on students' academic outcomes (Durlak et al., 2011; Cipriano, Strambler, et al., 2023b; Cipriano et al., 2024; Taylor et al., 2017). However, these meta-analyses did not conduct analyses that allow for examining various types of academic achievement. The comprehensive meta-analysis study by Durlak et al. (2011) laid the foundation for educational research, practice, and policy about the benefits of SEL with scientific evidence. Durlak et al. (2011) reviewed 213 school-based universal SEL studies and examined the intervention effects under six domains (i.e., SEL skills, attitudes, positive social behavior, conduct problems, emotional distress, and academic performance). The synthesized effect was reported as $g = .57$ for the overall SEL outcome across the six domains and $g = .27$ for the academic achievement as a subdomain, both significant. The authors further reported an 11 percentile-point increase in student achievement as a result of participating in a high-quality SEL intervention. The practical significance of these findings is noteworthy. An 11 percentile-point gain in academic performance implies that a student initially performing at the median (50th percentile) could improve to approximately the 61st percentile following participation in a high-quality SEL program. This magnitude of improvement underscores the potential of SEL interventions not only to enhance social and emotional competencies but

also to yield measurable benefits in academic achievement, thereby supporting the integration of SEL into core educational practice. Although robust, Durlak et al. (2011) reviewed SEL research published through 2007, and as a result, findings from newly conducted SEL research need to be analyzed to understand if these effects are similarly expected contemporarily. Furthermore, the impact of a subdomain of academic outcomes requires additional moderator effects to account for program features, such as program duration.

More than a decade later a contemporary meta-analysis by Cipriano, Strambler, et al. (2023b) reviewed the recent universal school-based SEL studies available between 2008 and 2020, focusing on its overall benefits for students, and twelve domains of SEL effects (i.e., SEL skills, attitudes/beliefs, prosocial behaviors, externalizing behaviors, civic attitudes/behaviors, peer relationships, emotional distress, school functioning, disciplinary outcomes, school climate/safety, family relationships, and physical health outcomes) from 258 universal SEL studies ($g = .194$). Cipriano, Strambler, et al. (2023b) reported that students who experienced SEL programs demonstrated significant improvements in their school functioning ($g = .12$)—a global indicator of school success—and discretely in students' academic achievement (i.e., one of the subdomains of school functioning outcomes; $g = .11$). However, this study did not analyze differentiation in types of student achievement results from the corpus of studies.

Similar findings were reported among articles assessing prior meta-analyses that examined SEL programs (Durlak et al., 2022; Greenberg, 2023). Among 12 meta-analyses comprising more than 500 studies representing approximately a million students worldwide, Durlak et al. (2022) found that less than a fifth of studies reported outcomes pertaining to academic achievement. The authors further found that among the 12 meta-analyses, four reported significant effects for academic achievement, and three explored follow-up effects, each of which reported significant effects for academic performance (Durlak et al., 2022). Similar results were reported by Greenberg (2023), who found effect sizes ranging from .18 to .28 across the four meta-analyses examining academic outcomes. Both Greenberg (2023) and Durlak et al. (2022) emphasized the consistency across findings, each highlighting the fact that four independent research teams, spanning seven years, all reported similar results: that students participating in universal SEL interventions saw significant improvement in their academic outcomes.

SEL and Academic Achievement

Studies examining whether SEL interventions improve academic achievement have found mixed results, with some indicating significant effects while others show no changes between intervention and control groups. For example, some research indicates that effects of SEL interventions on academic achievement vary across elementary (Aasheim et al., 2019; Berry et al., 2016; Chuang et al., 2020; Fishbein et al., 2016), middle (Paricio et al., 2020), and high (Dowling et al., 2019; Shoshani et al., 2016) school students. In general, the literature suggests that such variations are likely—in part, due to the age groups of the students (elementary, middle, and high school), how academic outcomes are assessed (e.g., test scores and GPA), the SEL program's length, and the content areas that are being assessed (e.g., math, reading).

Research suggests that school-based intervention effects on overall academic achievement may vary as a function of which SEL program and which academic outcomes are measured within a study. Among high school students in Ireland, Dowling et al. (2019) found that students participating in *MindOut* had improved SEL skills compared to students in the control group, but did not report significant differences in student scores on the School Achievement Motivation Rating Scale. In contrast, Shoshani et al. (2016) examined grade point averages among Israeli high school students who did and did not participate in the *Maytiv Program*, a positive psychology intervention, and reported modest differences between the groups. It is plausible that differences in findings can be attributed to the type of program being implemented as well as how the authors examined academic achievement. Additionally, findings may also be impacted by the length of each program, as *MindOut* was implemented across 13 weeks, while the *Maytiv Program* consisted of 15 bi-weekly sessions.

Different intervention effects on academic achievement have also been found between studies implementing the same SEL program across similarly aged students. Fishbein et al. (2016) compared academic achievement, using the Academic Competence Evaluation Scales (ACES), between elementary school students who were and were not exposed to the *PATHS* curriculum for one year and found that students in the intervention group reported significantly higher academic skills. Berry et al. (2016), on the other hand, examined the impact of a two-year *PATHS* curriculum on academic achievement of students in England using the *PATHS* Teacher Rating Scale (PTRS), which includes an academic performance subscale. Unlike Fishbein et al. (2016), the authors reported no significant differences in academic performance between groups after years one and two (Berry et al., 2016). Although there are a number of factors that could help explain why the *PATHS* curriculum had different effects on academic achievement across these two studies (e.g., implementation quality and student characteristics), the findings suggest that the type of measure used and how that tool evaluates student academic performance may also be contributing factors.

Research has shown varied effects of SEL programs on different subject matters with some studies reporting improved academic performance across each subject matter that was being evaluated (Lemberger et al., 2015; Weis et al., 2015), differing effects between subjects (Bakosh et al., 2016), or no effects at all (Diperna et al., 2016; Frank et al., 2017). Lemberger et al. (2015) and Weis et al. (2015), for example, examined reading and mathematics outcomes in elementary school and middle school students and found significant improvements among the intervention group in both subject areas. Conversely, Frank et al. (2017) found that English and math grades among middle school students did not differ between students who participated in an SEL-based yoga program and the control group. Diperna et al. (2016) reported similar findings among reading and math assessment scores between elementary school students who did and did not receive a positive classroom behavior intervention. Conversely, Bakosh et al. (2016) examined the impact of the *Mindfulness Based Social Emotional Learning* (MBSEL) program on the grades of elementary school students across six subjects: social studies, spelling, writing, math, reading, and science, and found significant intervention effects in reading and science. Research also indicates that focusing on

academic achievement among specific subject areas can overlook intervention effects on students' overall academic achievement. Among middle school students receiving *Second Step*, Espelage et al. (2016) found no differences in reading and math test scores between intervention and control groups. The authors did, however, find clinically significant improvements in GPA scores in the intervention group with a "medium effect" (Espelage et al., 2016, p. 329). In other words, the rise in GPA scores was not statistically significant, but there were notable differences between groups.

Given the variability in the findings across the literature, a systematic analysis would better understand whether SEL programs generally improve academic achievement among K–12 students, or vary as a function of grade levels, outcome measures, overall academic outcomes across subject areas, and program duration.

Purpose of This Study

This study is a systematic review and meta-analyses of universal school-based SEL programs on grades 1–12 students' academic achievement. The present paper disentangles the construct of academic achievement with the aim of providing clarity for SEL program providers, researchers, and adopters as to how achievement domains (math, literacy, mixed subjects), type of achievement measures (proximal-distal, grades, test) are affected by SEL programming for students across their school career (grades 1–12). Drawing from the registered reports by Cipriano, Strambler, et al. (2023b), this secondary meta-analysis included 40 universal school-based SEL studies, reflecting 30 discrete SEL programs, 12 countries, 33,737 students, and 76 academic achievement effects. Based on the synthesized evidence, we analyzed the overall effect of universal SEL programs on students' academic achievement and explored potential moderating effects through coefficient tests based on the meta-regression model. The following research questions and hypotheses guided this study:

- (1) Do SEL programs improve student academic achievement? We hypothesized that SEL programs would significantly and positively improve student achievement (Cipriano, Strambler, et al., 2023b; Durlak et al., 2011).
- (2) Do SEL program effects on student academic achievement vary by the types of academic achievement measure (i.e., GPA and tests)? We hypothesized that the teacher-reported grades would yield higher effects compared to standardized test scores, because grades also include teacher perceptions of student behavior and social skills (Guskey et al., 2019).
- (3) Do SEL program effects on student academic achievement vary across different subject areas (i.e., literary, math, and mixed subjects)? We hypothesized that the mean effects for literacy outcomes would be higher than the mean outcomes for math or mixed subjects, because numerous studies of SEL interventions have demonstrated that SEL programs that include literacy dimensions, such as storytelling, writing, reading, and discussion, are associated with significant academic effects (Rimm-Kaufmann et al., 2007; Rivers & Brackett, 2010).

(4) Do SEL program effects on student academic achievement vary across different participants' ages (i.e., elementary, middle, and high school students)? We hypothesized that the mean effects for elementary school achievement would be higher than the mean outcomes for middle and high school students, because younger students have previously been identified as being more flexible in affective learning from explicit social interactions than adolescents (Durlak et al., 2022; Immordino-Yang et al., 2019).

(5) Do SEL program effects on student academic achievement vary across different program implementation durations (i.e., up to four months and more than four months)? We hypothesize that longer-duration SEL programs (more than four months) will result in higher mean academic achievement outcomes compared to shorter-duration programs (up to four months), based on the expectation that sustained participation in SEL supports the ongoing development of students' social-emotional and academic skills (S. Jones et al., 2017; Taylor et al., 2017).

Methods

The present study followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses Literature Search Extension) guidelines to ensure a standardized review approach to conducting systematic review and meta-analysis (Moher et al., 2015; Page et al., 2021). In this section, we outlined the data collection methods used in the following subsections: inclusion criteria, search strategy, data screening and extraction, and risk for bias. See our prespecified, peer-reviewed protocol to strengthen transparency through preregistration in the supplemental material for additional detail (Pigott & Polanin, 2020).

Inclusion and Exclusion Criteria

We included studies of universal SEL programs from January 1, 2008, through December 31, 2020, that provided adequate information for the calculation of effect sizes in the domain of academic achievement. We defined universal school-based SEL interventions in this study as interventions for students provided during school hours and within the school setting that target the development of social and emotional skills among all students in the classroom or school. Social, emotional, behavioral, and academic outcomes include those subsumed in CASEL's five competencies as well as immediate and long-term indicators of student well-being, including prosociality, academic performance, educational attainment, employment status, motivation, engagement, conduct, and emotional distress (Cipriano, Strambler, et al., 2023b).

The universal SEL studies were included by meeting the following criteria: (1) appeared in English in any country between January 2008 and December 2020, (2) designed with pre-post measures and intervention-control groups, (3) implemented for K–12 learners in universal school education settings, (4) included at least six SEL sessions in the classroom or lasted at least four months in a whole school approach, and (5) extracted academic achievement-related outcomes measured in universal SEL programs. In the final screening stage, this meta-analysis dataset included outcomes that assessed subject-related achievements based on

GPA or standardized test results as part of its final inclusion criteria. GPA and standardized test scores are commonly used measures of academic achievement and academic success in schools. Studies were excluded if they did not meet any of these inclusion criteria.

Search Strategy

The search strategies derived from the PRISMA checklist enhance credibility and reduce potential biases of the included studies (Rethlefsen et al., 2021). With the support of a university librarian, an iterative process was used to translate and refine the searches, and to maximize sensitivity, the formal search used controlled vocabulary terms and synonymous free-text words to capture the concepts of “SEL programs” and “school.” The search strategy was peer-reviewed by a second librarian using the Peer Review of Electronic Search Strategies Checklist (McGowan et al., 2016). Full search strategies are available on the open science website (see the supplementary codebook).

In a comprehensive search for a universal SEL program study database, we used multiple search engines, such as APA PsycInfo (Ovid), MEDLINE (Ovid), Education Resources Information Center (ERIC), Dissertations & Theses Global (ProQuest), and Web of Science databases. Additionally, we reviewed multiple unpublished research reports shared by SEL researcher groups, such as the databases of the American Educational Research Association (AERA), the CASEL, and the RAND Corporation.

From the initial search database ($n=41,002$), 31,257 abstracts were identified after removing duplicates ($n=9745$). In the abstract screening, 30,034 cases were excluded following the exclusion criteria; a research team reviewed 1,223 full manuscripts. In total, 258 SEL studies implemented in universal K–12 education settings met the selection criteria. To ensure the reliability of study exclusion at this phase, all articles were independently reviewed by two of the authors, and any conflicts were resolved through discussion. Authors reached “almost perfect” interrater reliability for both stages of screening (0.93%–1% for title and abstract, and 0.81%–0.98% agreement for full text; McHugh, 2012), resulting in 382 SEL articles screened for inclusion in the parent meta-analysis. Finally, all the first authors of studies were contacted while research team members conducted ancestral searches of the reference lists of each included article using the *Web of Science* and *Google Scholar* tools.

Through a multistep screening process, studies reporting on duplicate data were excluded. As a result, the parent meta-analysis was refined to 258 SEL intervention studies at the K–12 level that met all inclusion standards. From this set, we conducted additional screening to include only those studies measuring students’ academic achievement results via GPA or standardized tests, ultimately including 40 empirical studies in this analysis (see Table 2 and online appendix A2). The current manuscript includes 40 universal SEL studies, from which we pooled 76 effect sizes of students’ academic achievement outcomes ($k=76$, $n=40$) (Figure 1).

Coding Procedures

In the parent meta-analysis, twelve coders (two undergraduates, four graduate students, two postdoctoral researchers, and four current study authors) coded

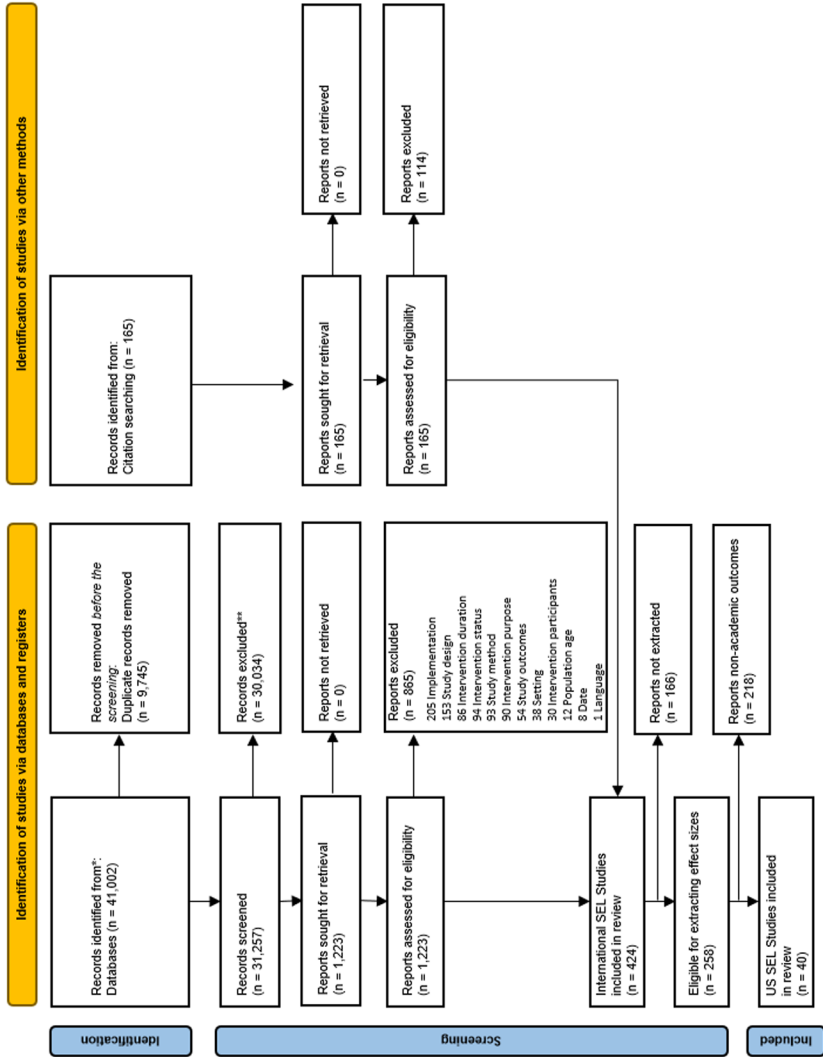


FIGURE 1. PRISMA diagram of included universal SEL studies in meta-analysis ($N = 40$, $k = 76$).

articles using Qualtrics (see full codebook from supplementary material for review). To effectively extract the large number of studies, and given the length of the codebook, three teams were established based on expertise to simultaneously extract study and student characteristics, intervention characteristics, and meta-analytic effects, respectively. All studies were double-coded, and given the heterogeneous nature of the body of literature on SEL, more than half of the studies were triple-coded to ensure reliability. To ensure coder consistency, calibration exercises were conducted before the review and weekly throughout full-text coding, which lasted nine months. Discrepancies were resolved through discussion between coders, and a third coder arbitrated when needed. Kappa coefficients corrected for chance agreement were acceptable across all codes reported in this review (mean kappa was .89). Raters' agreements on continuous variables were strong, all above .90, and agreement on categorical variables ranged from .78–.92, respectively. A full discussion can be found in Cipriano, Strambler, et al. (2023b).

This study aimed to analyze the effects of universal SEL interventions on students' academic success in K–12 settings. The meta-analytic dataset was restricted to studies reporting subject-related academic achievement using GPA or standardized test scores. This served as a core component of the inclusion criteria, and studies lacking objective indicators assessed by teachers or validated measurement tools were excluded from the analysis. To examine the heterogeneity of academic achievement outcomes and effects within the SEL empirical literature, extensive achievement outcomes were coded:

Academic achievement evaluation type: We coded for the type of academic achievement measure, including teacher-reported/assigned grade or grade point average (GPA) and standardized tests of achievement or related outcomes, such as the Woodcock-Johnson test. See Table 2 for a full list of achievement outcomes included in analyses.

Academic achievement subject: We coded for the content of the achievement outcome, including literacy, math, science, history, and/or mixed/aggregate subject content, etc. (subgroups: literacy, math, and mixed subject, and social science/science)

Student ages by school levels: We coded for the student ages based on the level of school where the SEL program was implemented. The codes included the elementary (5–10 years old; kindergarten through grade 5), middle (11–13 years old; grades 6 through 8), and high (14–18 years old; grades 9 through 12) schools. Although our sample included 12 universal SEL programs for 5-year-old students (kindergarten), none of these programs reported academic achievement outcomes for inclusion in our present analysis.

Duration: Programs were coded for length and explicitly for the number of sessions. For session-based programs, programs were organized into two groups: the programs that lasted less than four months but included more than six sessions and the programs that were longer than four months. For whole school (non-classroom-based programs), the duration was organized into two categories: up to four months in length or longer than four months in length. Programs under five sessions were excluded from the parent dataset to separate the effects of programs versus strategies or kernels (S. Jones et al., 2017).

Meta-Analytic Method

Power Analysis and Reported Effect Sizes

Analyzing at least 10 pooled results is generally recommended because meta-analytic studies should present synthesized results with statistically sufficient explanatory power (Valentine et al., 2010). We applied the calculator provided by Quintana and Tiebel (2019) to assess statistical power that estimates power based on key parameters, including the mean effect size, average sample size per study, total number of effect sizes, and the level of heterogeneity. This study included 40 SEL studies, reflecting the experiences of 33,737 students; the average number of participants in each study was more than 843 students. With 76 effect sizes and a high heterogeneity setting, the statistical explanatory power was .99, above the recommended threshold of .80 for reliable results.

In educational intervention research, comparing two groups (e.g., intervention and control) is the most common study design for examining the effectiveness of learning interventions (Ha, 2023a); the meta-analytic results from the pretest-post-test-control group design can provide a reliable synthesis of reviewed studies (Morris, 2008). In this study, we applied multiple adjustment processes of the effect size to provide concise meta-analytic results based on the current data structures; the following formulas show how we derived the reported effect sizes in this study. First, we calculated the adjusted Cohen's d from the selected studies within the pre-post control design. Effect size d was primarily calculated using mean and SD , as shown in the following formulas, with t -values and F -values utilized only in cases of limited result reporting. This approach compensates for the potential problem related to the lack of pre-test equivalence (Durlak, 2009). Next, we converted the effect size from the mean difference effect size d to g derived from Hedges' g (Hedges & Olkin, 2014); the converted effect size benefits to compensate for non-negligible bias related to small sample size issues (Cooper et al., 2019; Lin & Aloe, 2021).

- $S_{\text{pooled}} = \sqrt{\frac{(nt-1)St*St + (nc-1)*Sc*Sc}{nt+nc-2}}$
- Cohen's $d = \frac{Meant - Meanc}{Spooled}$
- Adjusted Cohen's $d = \text{Cohen's } d_{\text{post}} - \text{Cohen's } d_{\text{pre}}$
- $J = \left[1 - \frac{3}{4df-1} \right]$
- Reported Hedges' $g = \text{Adjusted Cohen's } d \times J$

In addition, we also used the variance of the effect size distribution to identify and adjust for outliers that could hinder the obtaining of precise meta-analysis results. We adjusted these effect sizes for potential outliers outside the 2 SD range to a maximum of ± 2 SD points (Cipriano, Strambler, et al., 2023b; Cipriano et al., 2024). We identified three effect sizes in this dataset that were outside the right side of the distribution and adjusted these values at the +2 SD point level. The

variances and standardized errors were recalculated based on Hedges' g within the adjusted settings. In this study, the reported effect size g suggests that a positive value reflects improved academic achievement in the SEL intervention group compared to the control group. Furthermore, as shown in the previous formulas, these effect sizes incorporate weights based on sample size, and their statistical significance can be assessed through 95% confidence intervals and p -values. The extracted effect sizes are detailed in Table 1, with further information about the confidence intervals available in the forest plot.

Meta-analytic Model

We analyzed the data using the robust hierarchical meta-analysis model, considering the clustered data structure and potential outliers. We applied the correlated and hierarchical effects (CHE) model, including the robust variance estimation (RVE) method, which derives the meta-analytic results considering potential correlations (Pustejovsky & Tipton, 2022). The analysis methods produce results that are more accurate than meta-analysis by accounting for correlated effect sizes derived from the same intervention (Cipriano, Strambler, et al., 2023b; Tipton & Pustejovsky, 2015).

In the overall meta-analysis, we applied the meta-regression model from the CHE and RVE methods using multiple packages in software *R*, such as *metafor*, *robumeta*, *meta*, *robust*, *dmetar*, *ggplot2*, and *clubSandwich*. Following the methodological guidance by Tanner-Smith and Tipton (2014), we conducted a sensitivity analysis assuming Rho (ρ) values of .40, .60, and .80. Considering our pilot results, we set the ρ -value as .60 in the final model and applied a new weighted variance derived from our clustered data. We reported the results estimated by a sandwich estimator based on the robust hierarchical meta-analysis approach (Tipton & Pustejovsky, 2015). To examine effects under varying conditions of SEL interventions, we conducted a moderation analysis and reported the slope differences among the embedded subgroup categories. We tested four moderation models (i.e., evaluation types, academic domains, school levels, and program durations) and reported the results of the coefficient test using the cluster-robust variance estimation-type 2 (CR2) method based on Satterthwaite's approximation of degrees of freedom. These techniques perform well with small sample sizes and correct degrees of freedom based on the variability and structure of the data. This results in conservative standard errors and more accurate p -values and confidence intervals, thereby reducing the risk of Type I errors.

Results

Characteristics of SEL Implementation

We identified 76 academic achievement effect sizes from 40 universal SEL studies conducted in K–12 educational settings ($n=40$, $k=76$). A systematic review included coding for specific SEL interventions, implementation duration, participants' school levels, sample sizes, achievement constructs, and measurements (see Table 1). The sample reflected 30 SEL programs conducted across 15 different North America, Asia, Africa, and European countries (i.e., the United States, Canada, Australia, China, Tanzania, England, Finland, Ireland, Italy, Israel,

TABLE 1*Characteristics of Universal SEL Implementations*

Study-level characteristics	<i>N</i>	Percentages (% of <i>N</i> =40)
Study Context		
US study	25	62.50
Outside of the US	15	37.50
Evaluation Types		
Grade point average (GPA)	14	35.00
Tests	27	67.50
Learning Outcome Contents in SEL		
Literacy	23	57.50
Math	19	47.50
More than two content areas, mixed subjects	18	45.00
Social Science	1	2.50
Science	1	2.50
School Levels (participants' ages)		
Elementary	27	67.50
Middle	6	15.00
High	7	17.50
Program Durations		
Up to 4 months with more than six lessons	20	50.00
More than 4 months	20	50.00
School Types		
Public	20	50.00
Public Charter	1	2.50
Alternative	1	2.50
N/A	18	45.00
Study Design		
Randomized controlled trials	29	72.50
Quasi experimental design	11	27.50
Peer-Review Study		
Yes	30	75.00
No	10	25.00
Funding Research		
Federal	15	37.50
State	1	2.50
Private foundation	4	10.00
University	3	7.50
No funding	16	40.00
Unknown	6	15.00

TABLE 2
Universal SEL Characteristics at Study-Level (N = 40)

Study Author	Intervention	Country Implemented	Intervention Duration (min./session)	School Level	Participants		Academic Constructs (Hedges' g): Measures
					Intervention	Control	
Aasheim et al. (2019)	Incredible Years	Norway	Up to four months (-)	Elementary	577	637	-Overall academic performance (.13): TRF
Antúnez et al. (2020)	SPIRALS	Spain	Up to four months (-)	Elementary	63	56	-Overall academic performance (-.18): The Woodcock-Johnson III Tests of Achievement
Bakosh et al. (2016)	MBSEL	USA	Up to four months (10)	Elementary	93	98	-Reading comprehension (.76): PROLEC-R -Social studies (-.15): grades by teachers -Spelling (-.04): Grades by teachers -Writing (.12): Grades by teachers -Math (.20): Grades by teachers -Reading (.37): Grades by teachers -Science (.46): Grades by teachers -Literacy (.26): MAP -Math (.16): MAP
Benson (2017)	Second Step	USA	More than four months (40)	Elementary	210	119	-Overall academic achievements (.16): Grades of five subjects (i.e., Kiswahili, English, math, science, and religion)
Berger et al. (2018)	ERSAE-Stress-Prosocial (ESPS)	Tanzania	Up to four months (90)	Middle	95	88	-Overall academic performance (.00): The PATHS Teacher Rating Scale (PTRS)
Berry et al. (2016)	PATHS	England	More than four months (25)	Elementary	2382	2177	-Overall achievement (.10): Grades by teachers -English (.13): grades by teachers -Math (.15): grades by teachers
Carroll et al. (2020)	KooIKIDS	Australia	Up to four months (-)	Elementary	557	292	-Reading (.04): The Woodcock Johnson Achievement Battery (WJ-III ACH)
Chuang et al. (2020)	Incredible Years	USA	More than four months (-)	Elementary	900	917	-Math (.05): The Woodcock Johnson Achievement Battery (WJ-III ACH)

(continued)

TABLE 2 (continued)

Study Author	Intervention	Country Implemented	Intervention Duration (min/session)	School Level	Participants		Academic Constructs (Hedges' <i>g</i>), Measures
					Intervention	Control	
Coates (2017)	Growing Early Mindsets	USA	Up to four months (-)	Elementary	147	179	-Oral reading fluency (-.01); DIBELS
Diamond et al. (2019)	Tools of the Mind	Canada	More than four months (-)	Elementary	172	180	-Reading (.65): A standardized tool-BC's objective -Writing (1.80): A standardized tool-BC's objective
Diperna et al. (2016)	Social Skills Improvement System Classwide Intervention Program (SSIS-CIP)	USA	Up to four months (25)	Elementary	210	192	-Reading (-.11): STAR Computerized test (Renaissance Learning, 2009, 2010) -Math (-.06): STAR computerized test (Renaissance Learning, 2009, 2010)
Diperna et al. (2018)	Social Skills Improvement System Classwide Intervention Program (SSIS-CIP)	USA	Up to four months (25)	Elementary	341	355	-Reading (.04): STAR Computerized Test -Math (.00): STAR Computerized Test
Donato (2009)	TestEdge	USA	Up to four months (30)	Elementary	41	41	-Math (.16): The Scott Foresman-Addison Wesley Math Diagnosis & Intervention
Dowling et al. (2019)	MindOut	Ireland	Up to four months (-)	High	295	243	-Overall school achievement (-.05); SAMRS
Espelage et al. (2016)	Second Step	USA	More than four months (50)	Middle	47	76	-Overall academic GPAs (1.78): GPA -Reading (.86): Illinois Standards Achievement Test (ISAT) -Math (.25): Illinois Standards Achievement Test (ISAT)

(continued)

TABLE 2 (continued)

Study Author	Intervention	Country Imple- mented	Intervention Duration (min./session)	School Level	Participants		Academic Constructs (Hedges' <i>g</i>): Measures
					Intervention	Control	
Felver et al. (2019)	Learning to BREATHE	USA	Up to four months (48)	High	12	11	-Overall academic grades(-.18): a 100-point scale GPA
Fishbein et al. (2016)	PATHS	USA	More than four months (20)	Elementary	164	164	-General academic skill(.29): Academic Competence Evaluation Scales (ACES)
Frank et al. (2017)	Transformative Life Skills (TLS)	USA	Up to four months (30)	Middle	80	79	-English (-.27): GPA -Math grade (.25): GPA
Gonzalez et al. (2020)	Tools for Life	USA	More than four months (-)	Elementary	1329	1329	-Reading and language arts (.02): MAAP RLA -Math (.00): MAAP math
Hanson et al. (2012)	Lessons in Character	USA	More than four months (20)	Elementary	507	507	-English language arts (.06): State English language arts assessments
S. M. Jones et al. (2010)	4Rs	USA	More than four months (-)	Elementary	515	427	-Overall academic skill (-.12): ECLS-K 1998 (National Center for Education Statistics, 2002)
Kokiniakis (2011)	Unnamed	USA	More than four months (50)	High	119	132	-Math (-.11): Operation Span Task (Ospan)
Lai et al. (2016)	The Little Prince is Depressed	China	Up to four months (60)	High	943	698	-Overall academic knowledge A (-.02) & B (-.17): Developed tests by authors
Lassander et al. (2020)	Healthy Learning Mind (HLM)	Finland	Up to four months (45)	Elementary	55	58	-Verbal fluency (.02): Trail Making test from D-KEFS CP (28-147) VF (39-130)

(continued)

TABLE 2 (continued)

Study Author	Intervention	Country Imple- mented	Intervention Duration (min./session)	School Level	Participants		Academic Constructs (Hedges' g): Measures
					Intervention	Control	
Lemberger et al. (2015)	Student Success Skills (SSS)	USA	More than four months (60)	Middle	111	82	-Reading (.38): Discovery Education Assessments (DEA) -Math (.54): DEA
McCormick et al. (2015a)	INSIGHTS	USA	Up to four months (120)	Elementary	225	210	-Reading (T1-T2) (-.19) & Reading (T1-T4) (.14): The LetterWord Identification and Applied Problems subtests Math (T1-T2) (-.11) & math (T1-T4) (.04): The Applied Problems subtest
McCormick et al. (2015b)	INSIGHTS	USA	More than four months (45)	Elementary	225	210	-Reading (-.06): The Woodcock-Johnson III Tests of Achievement (WJ-III) -Math (-.02): WJ-III
McCormick et al. (2016)	INSIGHTS	USA	Up to four months (120)	Elementary	84	220	-Reading (.28): WJ-III -Math (.28): WJ-III
Meadows (2018)	Unnamed	USA	Up to four months (30)	Elementary	75	43	-Reading (-.37): GPA -Math (-.34): GPA -ELA (.34): GPA
Meyer (2018)	Unnamed	USA	Up to four months (10)	Elementary	138	158	-Fluency (-.10): Standard reading assessment -Passages (RAPs) -Phonics (.03): RAPs
Moreno- Gomez and Cejudo (2019)	MindKinder	Spain	More than four months (15)	Elementary	48	26	-Listening vocabulary (.05): RAPs -Language comprehension (-.13): Children's Neuropsychological Maturity Questionnaire (CUMANIN) -Verbal fluency (-.12): CUMANIN -Writing (.03): CUMANIN -Expressive language (.21): CUMANIN -Reading (.22): CUMANIN -Verbal development (.76): CUMANIN

(continued)

TABLE 2 (continued)

Study Author	Intervention	Country Implemented	Intervention Duration (min./session)	School Level	Participants		Academic Constructs (Hedges' <i>g</i>): Measures
					Intervention	Control	
Mottram et al. (2016)	ReflectED	England	More than four months (30)	High	839	731	-Math (.27); CAS math
Muratori et al. (2016)	Coping Power	Italy	Up to four months (60)	Elementary	100	62	-Language & math (-.14); GPAs
O'Connor et al. (2014)	INSIGHTS	USA	Up to four months (45)	Elementary	183	162	-Reading (-.03); Academic Competency Evaluation Scale (ACES)
Paricio et al. (2020)	Unnamed	Spain	More than four months (50)	Middle	83	93	-Math (-.11); ACES -Academic critical thinking (-.10); ACES -Overall academic performance (.43); GPAs
Shoshani et al. (2016)	Mayriv	Israel	More than four months (90)	High	1262	1255	-School achievement (.29); GPAs
Stone (2009)	Olweus Bullying Prevention	USA	More than four months (-)	Elementary	3783	2008	-Academic achievement (.05); Measures of academic progress (MAP)
Weis et al. (2015)	PAX Good Behavior Game	USA	More than four months (-)	Elementary	523	426	Reading (.20); Measures of Academic Progress (MAP) Math (.27); MAP
Wright et al. (2010)	The Tai Chi Tiger Program	USA	More than four months (60)	High	62	56	-Overall academic grade (.15); GPAs

Notes. The references were listed in alphabetical order. N/A: not applicable.

Netherlands, Norway, and Spain), with the largest proportion of studies being conducted in the United States, 62.50% of the studies; $n=25$). *Incredible Years*, *INSIGHTS*, *PATHS*, *Second Step*, and *Social Skills Improvement System Classwide Intervention Program (SSIS-CIP)* were SEL programs that explored effects on achievement in multiple studies, whereas all other SEL programs were represented once in the sample (see Table 1). Of the included studies, 72.5% ($n=29$) were designed as randomized controlled trials and 27.5% ($n=11$) as quasi-experimental studies. Three-quarters of the included studies were peer-reviewed publications ($n=30$, 75.00%) compared to non-peer-reviewed reports and unpublished studies ($n=10$, 25.00%).

To address our research questions, we analyzed the type of academic achievement evaluation, academic domains, and the age of the students, alongside each study's characteristics. Regarding types of academic achievement, GPA-type academic achievement was reported in 14 studies (35.00%); standardized tests measured the test-based academic outcomes of students in 27 studies (67.50%). GPA, as an indicator of students' academic achievement, incorporates evaluated results by teachers and different school curricula. Although GPA tends to show high precision within a single school, its generalizability can be unstable in comparative studies across different schools or external groups (Bacon & Bean, 2006).

Students' academic achievement was measured primarily in literacy ($n=23$, 57.50%), math ($n=19$, 47.50%), and mixed subject learning outcomes from more than two content areas ($n=18$, 45.00%), with only one study (Bakosh et al., 2016) explicitly examining students' social science and science achievement ($n=1$, 2.50%). Regarding the ages of students served, studies were primarily in elementary schools ($n=27$, 67.50%), with middle and high schools representing 15.00% ($N=6$) and 17.50% ($n=7$) of the meta-analysis sample, respectively. Regarding the length of programs, studies were evenly split, with 20 SEL studies conducted for up to four months with more than six sessions, and the other 20 studies lasting more than four months.

Furthermore, we explored additional universal SEL program characteristics from the included studies (e.g., school types, funding support, and peer-review processes). Although some study characteristics could not be used for the subgroup meta-analysis due to an insufficient number of effect sizes in each subgroup or the considerable amount of inapplicable data coding (N/A), they provided sufficient descriptive results as a systematic review at each study level. For example, half of the included studies were conducted in public schools ($n=20$, 50.00%). In addition, 16 of the studies were supported by federal or state research funding ($n=16$, 40.00%), four by private foundations ($n=4$, 10.00%), and three by university-level funding ($n=3$, 7.50%). The presented information in Table 2 shows the additional systematic review results of students' academic achievement.

Results of Overall Meta-analysis

Publication Bias

We checked the publication bias of the overall sample using Egger's test (Egger et al., 1997). Considering the power and bias testing, we reported results for subgroups with at least 10 effect samples that allowed for statistical analysis. The included studies of this meta-analysis were collected from published and

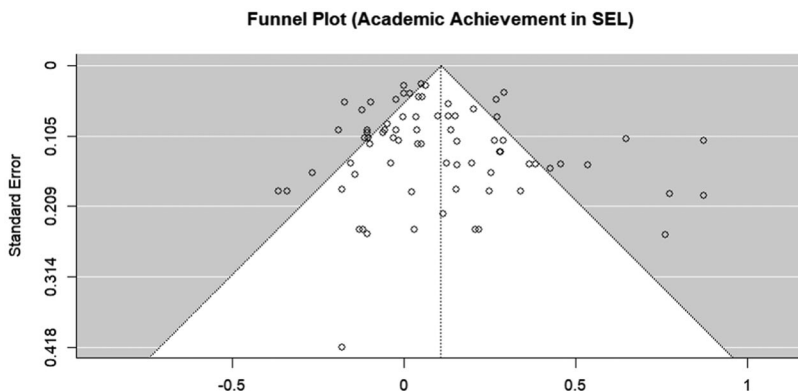


FIGURE 2. *Funnel plot of universal SEL studies in K–12 education ($k = 76$).*
Notes. Egger's $t = 1.02$, $df = 74$, $p = 0.309$.

unpublished papers from multiple search databases. Among 40 universal SEL intervention studies, we identified 30 peer-reviewed articles, seven thesis/dissertation papers, and three research reports. The proportion of unpublished studies was 25.50% in the final analysis dataset, and Egger's test result was not significant ($t [74] = 1.02$, $p = .309$). Additionally, we reviewed the distribution of the extracted effect sizes via funnel plots to identify potential biased cases and a symmetry contribution (Ha, 2023a); the funnel plot of the included universal SEL intervention studies did not show a severe asymmetrical shape (Figure 2). Taken together, we concluded that the current dataset did not show serious signs of publication bias. In the online appendix A1, we presented additional results for Eggers's t-test at each subgroup level.

Overall Effect of SEL on Academic Achievement

The universal SEL programs implemented across grades 1–12 exhibited a variety of implementation characteristics in the systematic review results (see Table 2), and heterogeneity tests revealed intervention effects to high heterogeneity ($Q [75] = 337.294$, $p < .001$; $I^2 = 85.64\%$). We evaluated the overall effect size pooled from all 40 universal SEL studies with various implementation characteristics. Given the highly heterogeneous nature of the data, we reported results with adjusted confidence intervals and p -values, considering hierarchical structures and robust variances. According to the *CHE* model results, K–12 students showed overall greater levels of academic achievement in universal SEL intervention groups compared to others in control groups with an effect size of $g = .101$ (95 $CI = [.035, .168]$, $p = .004$). The forest plot presents the visual summary of each intervention study included in the meta-analysis and school levels sorted by the individual studies' results (Figure 3).

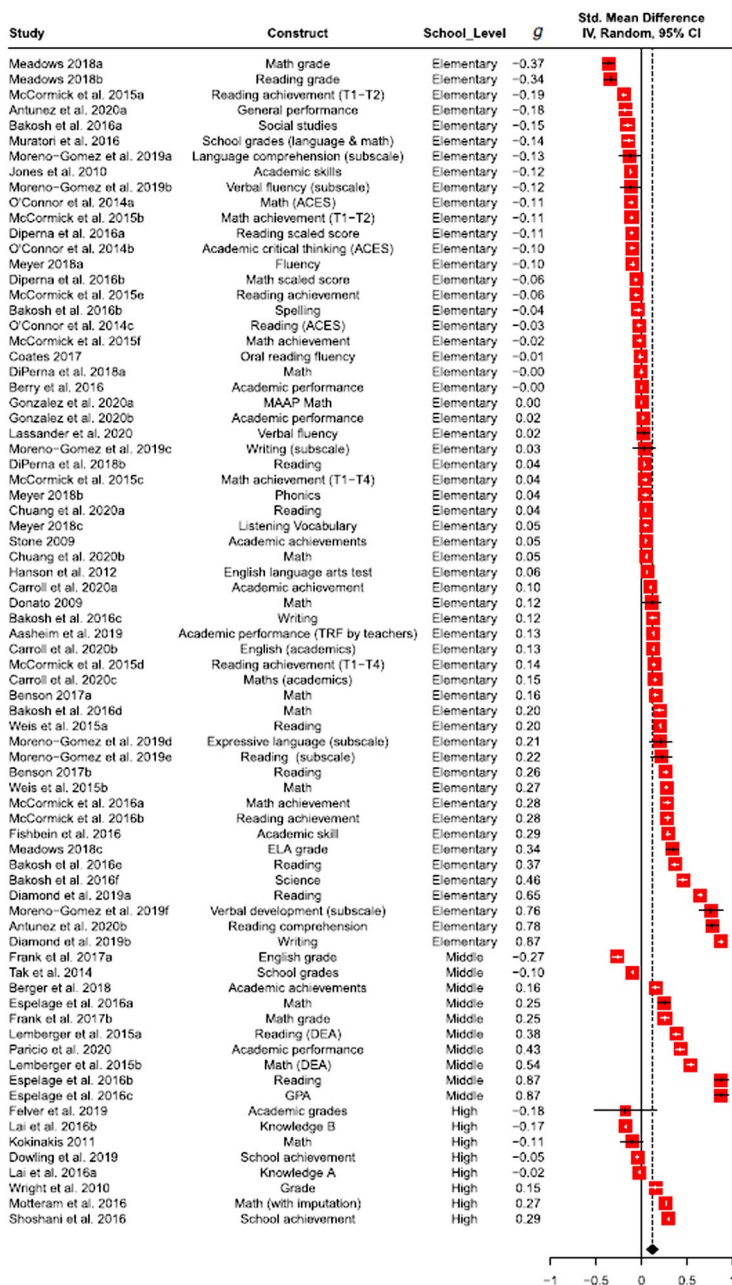


FIGURE 3. Forest plot of universal SEL studies in K-12 education ($N=40$, $k=76$; Hedges' $g=.101$).

Results of the Moderation Effects of the Meta-regression Model

In this study, the reported moderation analysis results were subjected to more robust testing by additionally applying the coefficient test function in meta-regression. Given that the k values for subgroups within the moderation model are smaller compared to the overall model, CR2 models were used to provide more reliable p -values and standard errors. This approach aims to examine effect sizes that respond more sensitively to subtle differences observed in program evaluation and implementation, making it suitable for exploring moderation model results. Therefore, it is important to note that the current findings were derived from a conservative analytical approach.

SEL Effect on Academic Achievement Moderated by Evaluation Types

We evaluated SEL effects on students' academic achievement by different evaluation types (see Table 3). The evaluation types were categorized as grades versus standardized test scores. We found 24 effect sizes from 14 studies that assessed student achievement with grades ($k=24, n=14$) and 52 effect sizes that measured student achievement by test from 27 studies ($k=52, n=27$).

Based on the meta-regression results with a moderator of evaluation types, the academic effect of SEL intervention measured by grade type outcomes was significant ($b=.111, se=.046, t=2.40, p=.038$). The other subgroup of test-type academic outcomes showed a statistically significant effect size ($b=.099, se=.041, t=2.38, p=.026$), and the effect sizes across different measurement methods were observed to be of similar magnitude. However, the effect of test-type had a smaller standard error and larger Satterthwaite degrees of freedom, suggesting that these results were estimated with more precision.

SEL Effect on Academic Achievement Moderated by Academic Domain

We examined the effects of the universal SEL programs on students' academic outcomes across subjects (see Table 4). Most universal SEL programs evaluated students' academic outcomes in literacy ($k=35, n=23$) and math ($k=20, n=19$)—additionally, the other 19 studies combined scores from multiple learning outcomes for the final report ($k=19, n=18$). While two studies reported the effects of the SEL program on social science and science achievement, they could not be included in this moderation effect analysis due to a limited sample size (Valentine et al., 2010). Therefore, the final results of the coefficient test were presented with three academic content domains: literacy, math, and mixed subject academic outcomes.

In the meta-regression analysis with the moderator of academic domains, we found the effect of SEL on literacy achievement to be statistically significant ($b=.144, se=.056, t=2.55, p=.019$). The effect of SEL on math achievement outcomes was also statistically significant ($b=.090, se=.040, t=2.23, p=.040$). In the model where we evaluated the universal SEL effect on mixed subject academic outcomes (i.e., across multiple combined academic subjects), we did not find a statistically significant result ($b=.056, se=.050, t=1.11, p=.283$).

TABLE 3

Universal SEL Intervention Effect on Academic Achievement Moderated by Evaluation Types

	<i>k</i>	<i>N</i>	<i>Coef. Estimate</i>	<i>SE</i>	<i>t</i>	<i>d.f. (Satt)</i>	<i>p (Satt)</i>
Evaluation Types							
Grades (GPAs)	24	14	.111	.046	2.40	9.61	.038*
Tests	52	27	.099	.041	2.38	22.21	.026*

Notes. *Satt*: Satterthwaite results from *RVE (CR2)* model;
**p* < .05.

TABLE 4

Universal SEL Intervention Effect on Academic Achievement Moderated by Academic Domain

	<i>k</i>	<i>N</i>	<i>Coef. Estimate</i>	<i>SE</i>	<i>t</i>	<i>d.f. (Satt)</i>	<i>p (Satt)</i>
Learning Contents							
Literacy	35	23	.144	.056	2.55	19.82	.019*
Math	20	19	.090	.040	2.23	16.64	.040*
Mixed subjects	19	18	.056	.050	1.11	14.48	.283

Notes. **p* < .05.

TABLE 5

Universal SEL Intervention Effect on Academic Achievement Moderated by School Levels

	<i>k</i>	<i>N</i>	<i>Coef. Estimate</i>	<i>SE</i>	<i>t</i>	<i>d.f. (Satt)</i>	<i>p (Satt)</i>
School level							
Elementary	58	27	.084	.036	2.317	22.11	.030*
Middle	9	6	.272	.132	2.052	4.77	.099 [†]
High	7	7	.057	.087	.065	4.26	.545

Notes. [†]*p* < .10 and **p* < .05.

SEL Effect on Academic Achievement Moderated by School Levels

Table 5 shows the effects of universal SEL on student academic achievement at different school levels. Effect sizes were analyzed for the three school levels (i.e., elementary, middle, and high school); none of the included intervention studies were implemented in kindergarten settings.

SEL programs implemented in elementary schools accounted for the largest proportion of effect sizes in our dataset ($k=58$, $n=27$), and the effect of SEL on achievement at this level was statistically significant ($b=.084$, $se=.036$, $t=2.317$, $p=.030$). Additionally, the moderation results showed no statistically significant results at the middle ($b=.272$, $se=.132$, $t=2.052$, $p=.099$) and high school level ($b=.057$, $se=.087$, $t=.065$, $p=.545$). With the data distribution showing a predominance of results from elementary schools and a smaller number of cases from middle and high schools, it is important to note that the p -values for the subgroups at these two school levels are derived from conservative estimates and might be reported as comparatively larger. For example, the aggregated mean effect size at the subgroup level was greater in the middle school group than in the elementary school group. However, the high heterogeneity observed in the distribution of effect sizes for programs targeting middle school students resulted in higher standard errors.

SEL Effect on Academic Achievement Moderated by SEL Program Durations

We analyzed 42 effect sizes from 20 SEL studies lasting less than four months with more than six sessions ($k=42$, $n=20$) and another 34 effect sizes from 20 SEL studies lasting longer than four months ($k=34$, $n=20$). Table 6 shows the moderation result based on the meta-regression model by different SEL program durations.

The effect of SEL on academic achievement was not statistically significant in studies with shorter intervention durations ($b=.026$, $se=.029$, $t=.917$, $p=.369$), and conversely, the SEL effect for the longer duration was statistically significant ($b=.185$, $se=.058$, $t=3.196$, $p=.005$). The results indicate that students who participated in SEL programs with a longer duration had significantly better academic benefits than those who were from controlled conditions (without SEL support in school curricula). Using the Satterthwaite degrees of freedom correction in the current moderation effect model was suitable to compare different program implementation conditions because the estimated p -values from the CR2 model tend to provide statistically conservative conclusions. The results derived from these tests indicate that student achievement is higher for students in longer-duration SEL programs than in shorter ones.

Discussion

The connection between students' social, emotional, cognitive, and academic development reveals how these dimensions work together to shape their overall human development in schools (Ha & Roehrig, 2022; Immordino-Yang et al., 2019; Zins et al., 2007). From a developmental perspective, it is expected that developing social and emotional competence during school-age youth will contribute to their successful academic outcomes (Farrington et al., 2012; Immordino-Yang et al., 2019; Osher et al., 2016). In the wake of the COVID-19 pandemic, there has been a growing need for the explicit instruction of social and emotional skills to help address diminished learning experiences across social, emotional, and academic domains among children and adolescents worldwide (IES, 2023; Rosanbalm, 2021). Yet, SEL programs, the organized curriculum interventions that explicitly teach social and emotional skills, are experiencing several

TABLE 6
Universal SEL Intervention Effect on Academic Achievement Moderated by Program Durations

	<i>k</i>	<i>N</i>	<i>Coef. Estimate</i>	<i>SE</i>	<i>t</i>	<i>d.f. (Satt)</i>	<i>p (Satt)</i>
Program Duration							
Up to four months	42	20	.026	.029	.917	14.8	.369
More than four months	34	20	.185	.058	3.196	16.6	.005**

Notes. ***p* < .01.

mounting challenges to implementation (Li et al., 2023; Shriver & Weissberg, 2020; Zieher et al., 2021). The following discussions offer educators and researchers critical discourse to inform our understanding of the effectiveness of school-based SEL for grades 1–12 students’ academic achievement. Based on the synthesis of available evidence, this study explored the variability in academic effects of SEL programs regarding different evaluation types, academic domains, ages of students, and program implementation durations.

School-based SEL Programs Improve the Academic Achievement of Students in Grades 1–12

First, our findings support that empirical intervention studies of school-based SEL programs across grades 1–12 demonstrate significant improvements in student academic achievement outcomes. The findings of this meta-analysis corroborate earlier research suggesting that fostering social and emotional competencies in education benefits students’ academic achievements, underscoring the essential role of schools in advancing integrated developmental goals (Darling-Hammond et al., 2022; Greenberg, 2023; Osher et al., 2020). Moreover, a key strength of this study lies in its advanced meta-analytic approach that applies the recently introduced analytical methods, leading to more precise statistical conclusions (see the methods section). These methods can reduce the overestimation of effects in traditional analytical models.

According to our results, between 2008 and 2020, studies exploring students’ academic achievements through SEL intervention programs were predominantly conducted in the United States, with many focusing on public elementary school settings. Regarding outcomes, these studies tended to emphasize changes in students’ literacy, math, and overall academic subjects. While numerous studies received support from various funding sources, it is notable that 40% of them were small-scale studies conducted without funding, suggesting the need for increased support to further investigate the scientific evidence of SEL effectiveness. In the recent global context, the OECD (2024) has been expanding its research on the impact of social and emotional skills on the academic success, active citizenship, and well-being of children and adolescent students in the post-COVID era. The OECD’s Survey on Social and Emotional Skills (SSES) project

underscores the increasing global awareness of the importance of social and emotional skills in adolescence (OECD, 2024).

SEL Improves Students' Grades and Achievement on Standardized Tests

Next, we synthesized a previously unexplored association between academic achievement and SEL, disentangling the effects of SEL on students' academic achievement by different evaluation types (i.e., GPAs and tests). Systematic review results revealed that SEL studies more frequently reported students' academic outcomes with standardized tests rather than GPA-based assessments. In the coefficient tests based on the meta-regression model, students in the SEL intervention group demonstrated higher academic achievement, as reflected in GPA and standardized test results, compared to those in the control condition, regardless of the evaluation types. The observed standard error of academic achievement was small in the test-type evaluation, indicating higher precision and a stronger central tendency around the mean effect size (Borenstein et al., 2009), along with a high Satterthwaite approximate degrees of freedom. These findings suggest that achievement test-based results (such as standardized tests) generated more accurate estimates of the effects of SEL on academic achievement. Overall, GPA was also higher for students in the intervention group compared to controls.

The results from GPAs indicate the variation in students' achievement by categorical intervals, which might have potential limitations in explaining the subtle differences among students ranked in the same grade. Furthermore, grades are reported by teachers, and although considered aggregates and reflective of student academic outcomes, they are more distal to student direct outcomes relative to student achievement test scores. Our findings suggest that using achievement measurement tools based on a test type with high reliability has benefits for examining the more precise impacts of SEL on students' academic outcomes than student grades.

Student Achievement in Literacy and Math Was Improved After Participating in SEL

Furthermore, we meta-analytically compared the academic effects of SEL on whether the program effects differ across various academic domains. The synthesized effects were presented descriptively in literacy (57.50%), followed by math (47.50%) and overall achievement from multiple academic domains (more than two academic contents from literacy, math, etc.; 45.00%). Students who participated in SEL programs demonstrated significantly greater achievement in both literacy and math compared to those in the control condition. However, in academic achievement results that combine two or more subject areas, the overall effect size was not statistically significant.

Our findings have notable implications for SEL implementation and anticipated effects on different academic domains. For example, among various SEL programs integrating social and emotional competence in subject learning, many empirical studies have been examined with aspirations of students' literacy achievement, and indeed, these programs demonstrated better academic outcomes than other subject domains (Benson, 2017; Rimm-Kaufman & Chiu, 2007; Rivers et al., 2013). In this study, synthesized results from various SEL programs on

math achievement revealed that despite SEL programs not directly teaching mathematical skills, students who participated in SEL programs demonstrated significantly better achievement in math compared to the control group. Across the various universal SEL programs span, the current study aligned with the previous meta-analysis study results that SEL programs positively contribute to students' better academic achievement (Cipriano, Strambler, et al., 2023b; Durlak et al., 2011). In evaluating students' academic outcomes as an indicator of mixed content areas, there is a potential for higher levels of heterogeneity compared to a singular subject domain. It is possible that students' academic benefits may vary by subject; we recommend researchers refine their measurement tools to enable the most accurate evaluations of how SEL contributes to student academic success in school.

The Effects of SEL on Academic Achievement Are Stronger at Elementary Level

We explored whether the academic effects of SEL programs vary according to students' ages, particularly across various school levels (elementary, middle, and high school). Programs for elementary school students were the largest proportion of studies in our sample, comprising about two-thirds of the studies (see Table 1). The synthesized results indicate that elementary school students who participated in the SEL intervention group showed statistically significant academic achievement compared to their counterparts in the control group. Notably, the Satterthwaite approximate degrees of freedom for elementary school students' academic achievement changes were observed to be considerably higher than at other school levels. This reflects the low variance among the observed cases, supporting the effect size stability and significance across the studies.

The academic results from middle school students in SEL programs were more heterogeneous, and the range of effect sizes was wider than for elementary school programs. We also found that Egger's test result of middle school programs detected a potential asymmetry issue among the included studies. For instance, the average effect size of aggregated academic achievement among students was larger for middle school programs compared to elementary school programs. However, due to the high heterogeneity among the studies, the *p*-value calculated using a conservative coefficient test ultimately determined that this trend was not statistically significant. This contribution might be related to small sample studies; therefore, more empirical studies with larger samples would be needed for further exploration. Unfortunately, too few empirical studies examined high school students' academic effects from SEL programs to allow for inclusion in our analyses. We suggest future research to gather more evidence to elucidate the concise academic effects of SEL for students in middle and high schools, respectively. In addition, the smaller variance in the effect size results of academic achievement among elementary school students participating in SEL programs compared to those of middle and high school students is evidence of this. Researchers highlighted that youth and adolescents' self-reflection and positive interactions with peers can increase their engagement in learning; it is also related to their successful academic outcomes (Darling-Hammond et al., 2022; Domitrovich et al., 2022). In particular, Ha and Cipriano (2025) found a pattern in a three-year longitudinal study tracking the relationship between students'

social-emotional factors and their academic achievement, where the connection between these factors became stronger as the students' school grade levels increased; this means that the positive impact of social-emotional skills grows gradually over the developmental processes. The academic success experiences in secondary schools may lead to successful college or workplace adaptation later on. Also, such skills are not naturally acquired in adulthood but are progressively improved through diverse educational encounters throughout schooling (Greenberg et al., 2003; Immordino-Yang et al., 2019). Therefore, offering scientific evidence through empirical investigation would be essential to aid positive evolutions throughout the whole human development stage (i.e., from childhood to adulthood) beyond the school ages.

Longer SEL Experiences Have Significant Effects on Student Academic Achievement

We report a new understanding of how students' academic effects vary depending on the program intervention durations. Students who participated in SEL programs that lasted longer than half a school year had significantly better academic achievement outcomes than students in the control condition. An intriguing finding of this study is that the academic achievement benefits for students in intervention programs lasting less than one semester were not statistically significant. If the interest is to strengthen students' academic outcomes, then results imply that longer investments in students' SEL skill development are optimal. As an example, McCormick et al. (2015a) conducted a study tracking changes in students' academic achievement under short-term and long-term interventions, showing that longer intervention periods result in more positive and statistically significant effect sizes. Given that most SEL programs (88.10%) are not explicitly focused on improving academic outcomes for youth (Cipriano, Strambler, et al., 2023b), our findings suggest that ensuring that students have developed a strong foundation of SEL skills through their experience in their SEL program across the whole school year makes a statistically significant difference in their academic achievement at the conclusion of the program.

Results reinforce the idea that positive academic achievement outcomes can be anticipated with programs with a longer-term duration (Taylor et al., 2017). Furthermore, in connection with our results, if the enhancement of grades 1–12 students' social, emotional, and academic is not an immediate change right after a short intervention, but manifests over the long term through steady and continuous SEL experiences, future SEL research needs to consider increasing its positive effects by creating a supportive developmental ecology where such students can learn social-emotional competencies in schools (S. M. Jones & Khan, 2017; S. M. Jones et al., 2019).

Practical Implications

Students' social-emotional skill development and academic achievement outcomes are the primary and essential outcomes that drive interventions for students and schools worldwide (OECD, 2023, 2024). The rationale behind universal SEL is based on the understanding that promoting students' social and emotional skill growth in school positively influences their academic outcomes (Weissberg,

2019). Literacy, above all other academic domains, is regarded as a universal currency for economic mobility and life success, followed closely by mathematical competence to support students and adults with fiscal management, technological advances, and industry. Although supporting the explicit social and emotional development of young people worldwide has gained increased attention and investments over the past three decades, and robust evidence speaking to its large-scale effectiveness is mounting (Cipriano, Strambler, et al., 2023b; Durlak et al., 2011, 2022; Greenberg, 2023), SEL programs continue to be positioned in schools worldwide as optional—nice to have but not critical for supporting student success in school (Schwartz et al., 2020; Yoder & Dang, 2023).

This study presents valuable evidence in social science studies through an updated systematic and meta-analysis review that thoroughly examines the effects and possibilities of SEL on students' academic growth, as addressed by scholars in various conceptual models from previous research (Greenberg, 2023; Osher et al., 2020; Weissberg, 2019). The significant effects we report here regarding students' tests and GPAs, and literacy and math achievement, are important and timely; academic recovery is of paramount importance both domestically (Lewis & Kuhfeld, 2023; National Academies of Sciences, Engineering, & Medicine [NASEM], 2023) and internationally (OECD, 2023).

Recent benchmarks in educational research suggest interpreting the magnitude of effect sizes as less than .05 is small, .05 to less than .20 is medium, and .20 or greater is large (see Kraft, 2020, for review). Effect sizes in these analyses ranged from .08 to .19, suggesting the effect of SEL intervention on student achievement, in general and in both literacy and math, across test types and for longer duration programs is moderate. Given the great deal of heterogeneity in interventions, implementation, and quality, findings serve as a strong estimate of favorable anticipated effects of SEL interventions on student academic achievement. Furthermore, to support the translation of the practical implications of these effects, we transformed effect sizes using the improvement index (What Works Clearinghouse [WWC], 2017), which is a commonly used model for explaining the expected change in percentile rank for an average comparison group student if the student had received the educational intervention (WWC, 2017). On average, our analyses suggest a 4.2 percentile point increase in an intervention group student's overall academic achievement, a 6.3 percentile point increase in an intervention group student's literacy achievement, and a 3.8 percentile point increase in an intervention group student's math achievement, after receiving an SEL intervention. Furthermore, students who participate in longer-duration SEL programs could expect an 8.4 percentile point increase in an intervention group student's overall academic achievement after receiving an SEL intervention. Given that these estimates are based on a highly heterogeneous sample of programs, achievement measures, and quality levels of implementation, we interpret these findings as conservative estimates of the smallest possible effects of SEL interventions on students' academic achievement and encourage educational researchers to continue to collect and report on indicators of quality and achievement to allow for further study and analysis.

Given the significant and robust effects SEL programs have on discrete indicators of academic achievement, students worldwide stand to benefit academically

from access to quality SEL programming. The science of learning and development has long demonstrated the interconnectedness of learner cognitive, affective, and behavioral skill development and its implications for academic functioning and life success (Immordino Yang et al., 2019; Osher et al., 2020). Once more, given the current state of diminished academic achievement (OECD, 2023) and the concurrent rising mental health concerns (UNICEF, 2021) for youth worldwide, the provision of explicit SEL instruction through the implementation of a universal school-based SEL program stands to address these disparaging trajectories with evidence-based practice. Indeed, cost estimates continue to affirm that investments in SEL are a strong return on investment (Belfield et al., 2015) across a myriad of social and economic experiences for youth and adults across their lifespan.

SEL can be the first step in helping students prepare to learn successfully because school-aged students learn from positive and diverse social interactions in school (Darling-Hammond et al., 2022; Phillippe, 2021; Weissberg, 2019). Previous empirical findings have shown significant interaction effects among students' social-emotional factors and cognitive skills on academic achievement growth across different subject areas (Ha, 2023b; Ha & Roehrig, 2022). Results suggest that students with stronger social and emotional skills can expect better academic growth when they are prepared for learning. Aligned with the current study's findings, these positive academic developments can be observed in SEL-rich environments. In light of the important role school education plays in nurturing the social, emotional, and academic growth of all learners, it becomes imperative for future educational policies to prioritize equitable access to universal SEL programs. Such inclusivity will contribute to providing equitable learning opportunities for all learners to support them in reaching their full potential and realizing holistic human development.

Limitations and Further Study

This review focused on examining the academic effects of universal school-based SEL programs embedded in the regular school day; therefore, a limitation of the current study is the exclusion of a number of SEL programs in other delivery types, such as out of school time (OST; Cottone et al., 2013; Hurd & Deutsch, 2017). In this research, we focused on evaluating the academic effects of SEL programs that are intended to be delivered and accessible to all students in schools, which we refer to as Tier 1 programs. We excluded other SEL programs that require individual student or parental selection, as these programs would have different aims and characteristics compared to the currently included programs. Based on the current findings, further research would benefit from expanding the inclusion criteria to examine additional delivery methods, such as OST programming, programs explicitly designed to address gaps in student achievement, and programs that are targeted as Tier 2 or 3 support for specific profiles of students.

This study's findings synthesize evidence from published or unpublished papers in English from 2008 through 2020. The included empirical studies are based on data from the pre-COVID-19 era. Post-2021, school closures and the shift to online learning have hindered both the implementation of educational programs in regular school settings and data collection (IES, 2023; Pressley & Ha,

2022; Zieher et al., 2021). Consequently, we recommend that future research focus on exploring SEL outcomes for students in a variety of educational settings and that new research questions explore the effects of SEL conducted post-COVID-19 in new review studies, particularly those focused on understanding and mitigating learning loss for school-age children.

In addition, the current study was limited to studies with full-text access available in English. Though this study included balanced US and non-US empirical studies, this approach implies that certain empirical studies from non-English-speaking countries may not have been fully included in our analysis due to language limitations. In the interest of equity of voice in educational research, future research should explore ways to ensure that the diverse research findings proposed by researchers from non-English-speaking countries can be broadly shared in the SEL research networks. Future meta-analytic teams exploring the efficacy of SEL programs should also consider partnering with researchers around the globe to allow for the inclusion of papers not written in English for analysis.

Further, analyses were limited to the information available across studies within our current dataset. The current study examined how evaluation type, learning domain (e.g., math), school level, and program duration moderated the SEL program effect on academic achievement. Given the lack of studies focusing on social science and science as a learning outcome, we were restricted to examining the moderation effects of only three of the five learning outcome contents represented across studies. Also absent from this analysis is a deeper examination of the role of school climate in shaping academic outcomes associated with SEL.

School climate serves as a critical contextual factor in SEL research, as it significantly influences both the fidelity of implementation and the sustainability of SEL interventions. The implementation of SEL programs is pivotal in addressing the multifaceted challenges that students face globally, including declining academic achievement, diminished social-emotional well-being, and persistent educational inequities. The broader school ecology and climate can exert lasting influence on student development at both micro- and macro-levels (Bronfenbrenner & Morris, 1998; McCormick et al., 2015a). A whole-school SEL approach that emphasizes environmental transformation seeks to foster foundational human development by cultivating a supportive climate for social, emotional, and academic growth (Cipriano, Naples, et al., 2023a; Hoffman, 2009). Over the past three decades, various school-based SEL initiatives have contributed to the establishment of positive school climates and the enhancement of students' social, emotional, and academic competencies (Weissberg, 2019). These programs operate under the premise that fostering a positive climate through SEL leads to improved mental health, more adaptive attitudes and behaviors, and greater academic success (Rimm-Kaufman & Chiu, 2007). Moreover, students' social and emotional development has been consistently linked to enhanced academic performance (Farrington et al., 2012; Osher et al., 2016). Future research would benefit from explicitly examining how school climate functions as a mediator or moderator in the relationship between SEL programming and academic achievement. Further, considerations for various aspects of intervention implementation were also limited, as this study only examined duration and dosage. To enable advances in programmatic design and implementation, SEL intervention research

must consistently measure for and report data on the quality of how the intervention was delivered and fidelity, as such aspects of implementation impact program outcomes (Cipriano, Strambler, et al., 2023b). Also missing from this study were subgroup analyses based on student characteristics. Future research would benefit from analyzing the academic effects of SEL programs for subgroups of students at the axes of salient learner identities (such as for students with learning disabilities or multilingual learners). Lastly, the present review does not address the follow-up effects of SEL interventions on academic achievement, and future research would benefit from an analysis of follow-up effects specifically on student achievement trajectories in the long term.

Conclusions

The findings of the current systematic review and meta-analysis elucidate the characteristics and impact of school-based universal SEL programs on the academic achievement of students across grades 1–12. The synthesized results demonstrate that SEL interventions within school settings significantly and positively contribute to students' successful academic achievement compared to the control condition. Our research provides evidence of SEL effects across specific types of achievement, academic domains, and age groups, using contemporary analytic models and statistically conservative methods. Notably, our findings demonstrate that SEL programs that are longer in duration produce more positive academic achievement outcomes for students than shorter SEL programs. While SEL programs are primarily intended to foster students' social and emotional development and well-being rather than explicit instruction in academic competencies, according to our results, there is robust evidence to suggest that SEL programs not only serve their intended primary purpose but also significantly increase students' academic success in schools.

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Data Availability

Relevant data and supplementary files presented in this manuscript are available on the Open Science Framework at <https://osf.io/6pqx2>. Additional data is available upon request to the corresponding author.

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***References marked with an asterisk indicate the study included in the meta-analysis.**

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